Estimating peer effects in noisy, low-rank networks via network smoothing

Alex Hayes

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Oregon State University

Roadmap for today

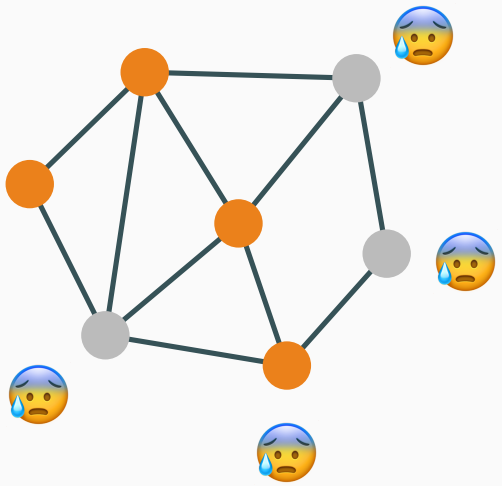
1. What are peer effects and how do we estimate them?
2. Peer effects in latent spaces
3. Shameless self-promotion

Peer effects

Humans constantly influence their social contacts



These social influences are often called peer effects

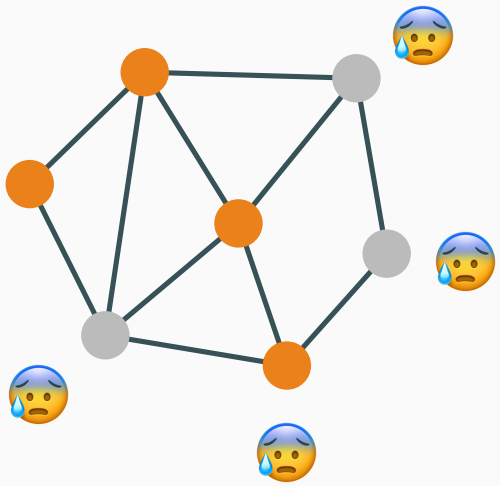


Direct effect: if I meditate, I am less likely to be anxious for my final 😰

Contagion: if my friends are anxious 😰, I am more likely to be anxious 😰

Interference: if my friends meditate, I am less likely to be anxious 😰

These social influences are often called peer effects



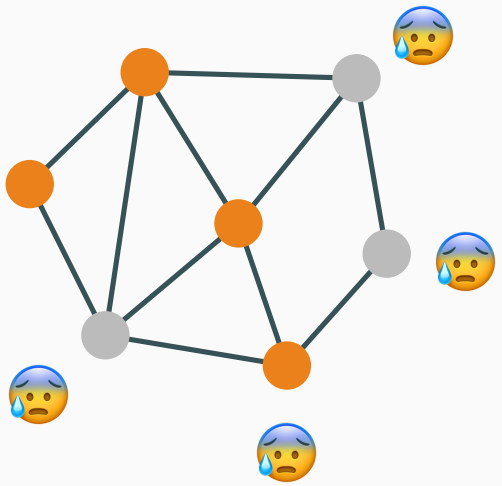
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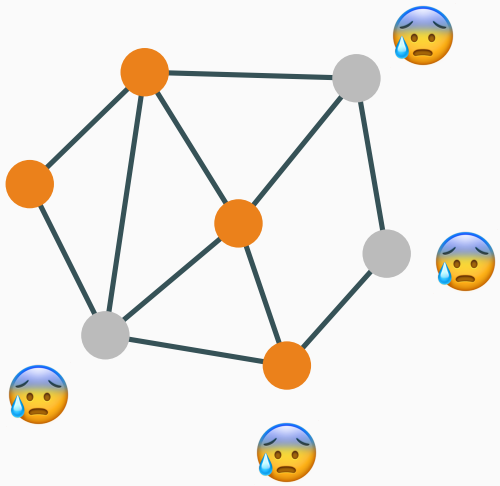
* Can be defined counterfactually, but we do not consider counterfactual inference in this talk.

Measuring someone's friends is challenging



If I ask you who your friends are:

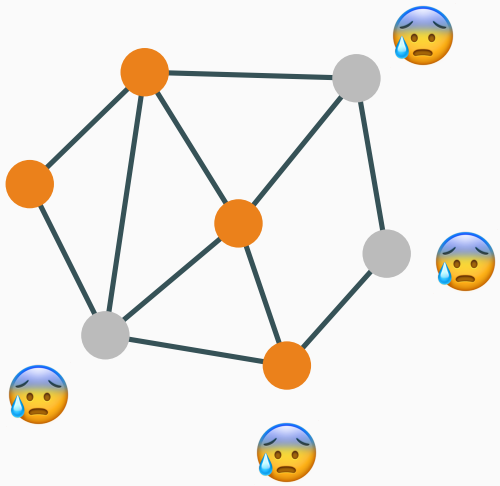
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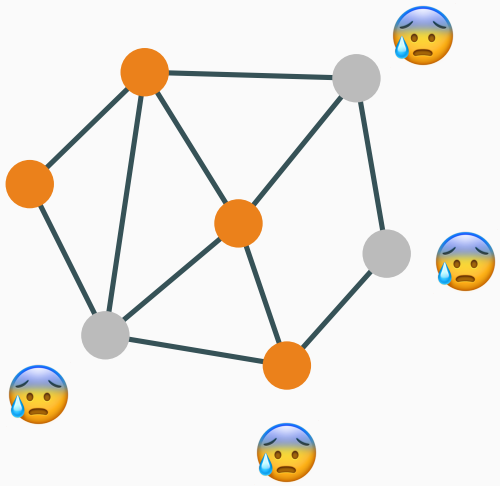
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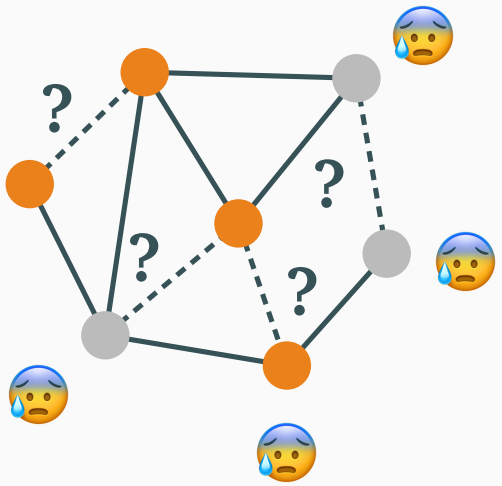
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If I ask you who your friends are:

- answers vary depending on how exactly I ask you,
- you'll probably forget day-to-day social contacts, and
- you might think someone is your friend, but they might disagree.

Measuring someone's friends is challenging



If I ask you who your friends are:

- answers vary depending on how exactly I ask you,
- you'll probably forget day-to-day social contacts, and
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Peer effects estimates need perfectly observed networks

$$Y_i = \beta_0 + W_i \beta_w + \underbrace{\beta_y \sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{peer contagion}} + \epsilon_i$$

…but the edges A_{ij} are always measured with noise!

Hold on! Does this model even make sense?

Write the model in matrix form, with $G = D^{-1}A$, such that $G_{ij} = A_{ij}/d_i$:

$$Y = \beta_0 1_n + W\beta_w + \beta_y GY + \epsilon$$

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$$Y = \left(I - \beta_y G\right)^{-1}\left(\beta_0 1_n + W\beta_w + \epsilon\right)$$

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\begin{aligned}
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Y &= \left(I - \beta_y G\right)^{-1}\left(\beta_0 1_n + W\beta_w + \epsilon\right) \\
&\overset{*}{=} \underbrace{\sum_{k=0}^{\infty} \beta_y^k G^k}_{\substack{\text{repeated} \\ \text{neighborhood} \\ \text{averaging}}} \left(\beta_0 1_n + W\beta_w + \epsilon\right)
\end{aligned}
$$

* Must have $\left|\beta_y\right| < 1$, so the effect of averaging decays with repetition

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\end{aligned}
$$

* Must have $\left|\beta_y\right| < 1$, so the effect of averaging decays with repetition

Because Y_i and Y_j are dependent, cannot use ordinary least squares → typically use two-stage least squares/instrumental variables instead

Peer effects estimates need perfectly observed networks

$$Y_i = \beta_0 + W_i \beta_w + \underbrace{\beta_y \sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{peer contagion}} + \epsilon_i$$

…but the edges A_{ij} are always measured with noise!

Peer effects in latent spaces

We propose a model based on probability of friendship instead

$$Y_i = \beta_0 + W_i\beta_w + \underbrace{\beta_y \sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{peer contagion}} + \epsilon_i \qquad \text{requires all } A_{ij}$$

$$Y_i = \theta_0 + \underbrace{W_i\theta_w}_{\substack{\text{observed} \\ \text{covariate} \\ \text{effects}}} + \underbrace{\theta_y \sum_{j \neq i} \frac{\mathbb{P}\left(A_{ij} = 1\right)}{\tilde{d}_i} Y_j}_{\text{latent contagion}} + \epsilon_i \qquad \text{requires } \mathbb{P}\left(A_{ij} = 1\right)$$

where $d_i = \sum_j A_{ij}$ is degree and $\tilde{d}_i = \Sigma_j \mathbb{P}\left(A_{ij} = 1\right)$ is expected degree.

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where $d_i = \sum_j A_{ij}$ is degree and $\tilde{d}_i = \Sigma_j \mathbb{P}\left(A_{ij} = 1\right)$ is expected degree.

Two big questions:

1. How to get friendship probabilities $\mathbb{P}\left(A_{ij} = 1\right)$?
2. Do the coefficients θ_0, θ_w and especially θ_y from the latent contagion model mean anything?

We estimate friendships based on a low-rank network model



Intuition: stochastic blockmodels

Block indicators Z_i

Popularity parameters ω_i

Mixing matrix $B \in [0,1]^{r \times r}$

$$\mathbb{P}\left(A_{ij} = 1 \mid Z, \omega, B\right) = \underbrace{\omega_i Z_i B Z_j^\top \omega_j}_{\text{rank } r}$$

Only assume undirected non-negative $A \in R^{n \times n}$ with low rank $P = \mathbb{E}[A \mid U, S] = USU^\top$ and sub-gamma (Bernoulli, Poisson, Gaussian, Gamma, …) deviations around A.

We estimate friendships based on a low-rank network model



Let $\widehat{U}\hat{S}\widehat{U}^\top$ be the rank r SVD of A.

Then $\left[\widehat{U}\hat{S}\widehat{U}^\top\right]_{ij} \to \mathbb{P}\left(A_{ij} = 1\right)$

Principal components of network!

Okay but does the latent contagion model mean anything?

$$Y_i = \beta_0 + W_i\beta_w + \underbrace{\beta_y \sum_{j \neq i} \frac{A_{ij}}{d_i} Y_j}_{\text{peer contagion}} + \epsilon_i \qquad \text{requires all } A_{ij}$$

$$Y_i = \theta_0 + \underbrace{W_i\theta_w}_{\substack{\text{observed} \\ \text{covariate} \\ \text{effects}}} + \underbrace{\theta_y \sum_{j \neq i} \frac{\mathbb{P}\left(A_{ij} = 1\right)}{\tilde{d}_i} Y_j}_{\text{latent contagion}} + \epsilon_i \qquad \text{requires } \mathbb{P}\left(A_{ij} = 1\right)$$

where $d_i = \sum_j A_{ij}$ and $\tilde{d}_i = \Sigma_j \mathbb{P}\left(A_{ij} = 1\right)$.

First: we can estimate the latent contagion coefficients

$$\text{Noisy } A \xrightarrow{\text{rank-}r \text{ SVD}} \hat{P} \xrightarrow{\text{row-normalize}} \hat{G} = \hat{D}^{-1}\hat{P} \xrightarrow{\text{2SLS}} \hat{\theta}$$

- **Identification:** instruments $\left[W\ \hat{G}W\ \hat{G}^2W\right]$ must add rank beyond W → requires $\hat{G}$ to vary non-linearly across vertices (via $\hat{D}^{-1}$) and r small relative to the number of covariates; "sufficiently dense" means expected degree grows fast enough that $\hat{P}$ converges.

Theorem: In sufficiently dense low-rank networks, $\hat{\theta}$ is asymptotically normal under the latent contagion model

$$\sqrt{n}\left(\hat{\theta} - \boldsymbol{\theta}\right) \to \mathcal{N}\left(0, \Sigma\right)$$

* Slightly simplified for intuition, need additional terms for identification

This is asymptotically as good as knowing exact friendship!

Theorem: $\hat{\theta}$ is asymptotically normal under the peer contagion model

$$\sqrt{n}\left(\hat{\theta} - \beta\right) \to \mathcal{N}\left(0, \Sigma\right)$$

Latent contagion is a reasonable working model when peer contagion is the true model

Asymptotically, β and θ have the same interpretation

Let $\tilde{d}_i = \sum_j P_{ij}$ and $\tilde{D} = \text{diag}\left(\tilde{d}_1, ..., \tilde{d}_n\right)$ and $\tilde{G} = \tilde{D}^{-1}P \in R^{n \times n}$.

Population design matrices:

$$X = [W\ GY] \qquad \text{peer contagion}$$

$$\tilde{X} = \left[W\ \tilde{G}Y\right] \qquad \text{latent contagion}$$

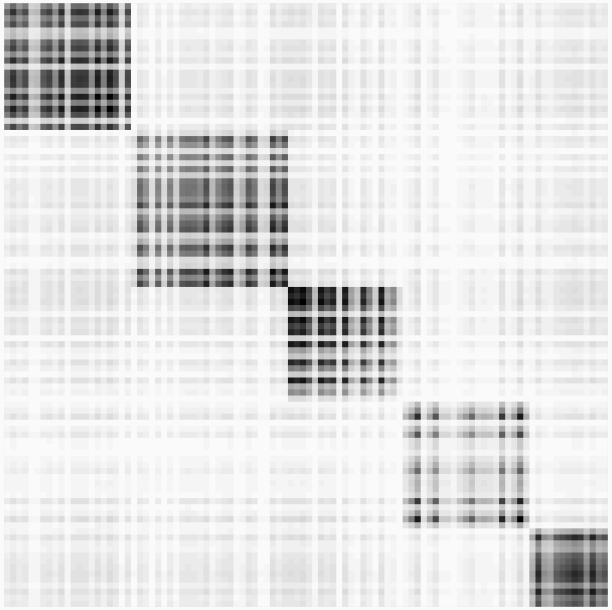
Theorem: Under both peer and latent contagion models, in dense low-rank networks, the non-parametric projection functionals are very close

$$\left\| \mathbb{E}\left[X^\top X\right]^{-1} \mathbb{E}\left[X^\top Y\right] - \mathbb{E}\left[\tilde{X}^\top \tilde{X}\right]^{-1} \mathbb{E}\left[\tilde{X}^\top Y\right] \right\| = o_p\left(\frac{1}{\sqrt{n}}\right)$$

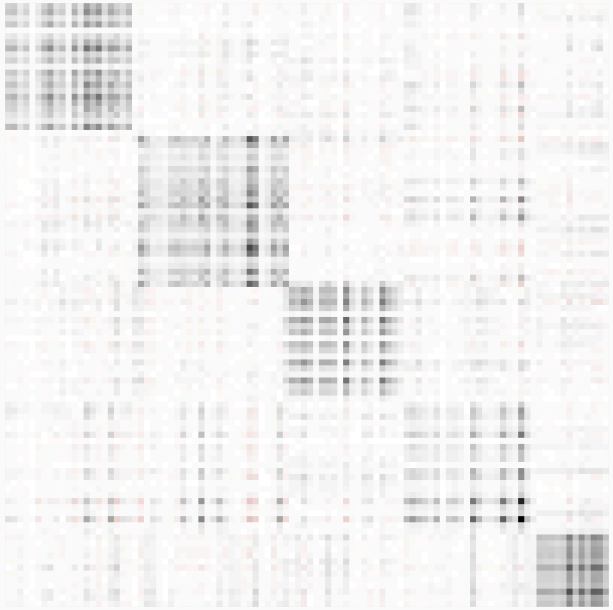
Why are latent contagion and peer contagion models so close?



A



$\mathbb{E}[A] = P$



$\hat{P}$

Estimating peer effects from noisy data

The latent contagion estimator handles noisy network data

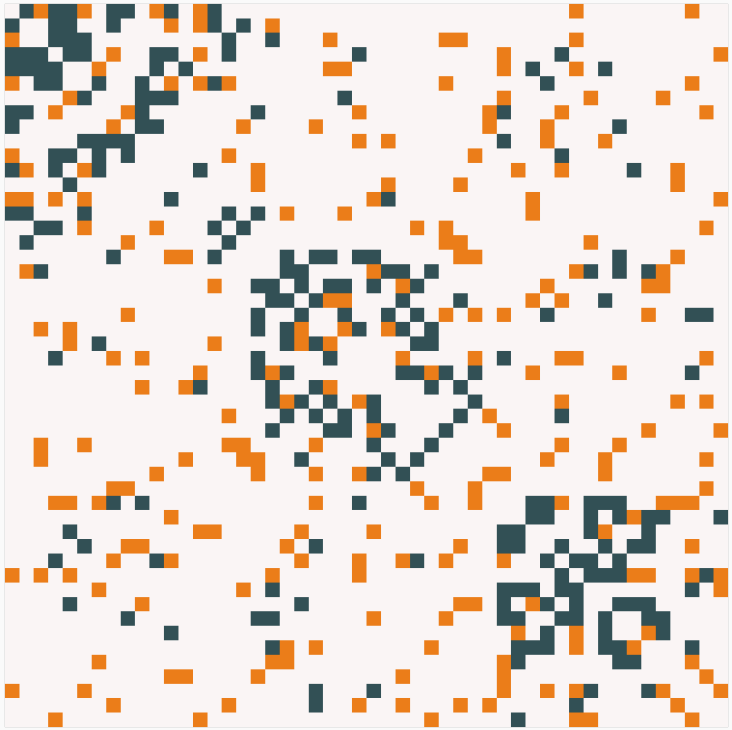
$\hat{\theta}$ only needs Y, W and an estimate $\hat{P}$ of P

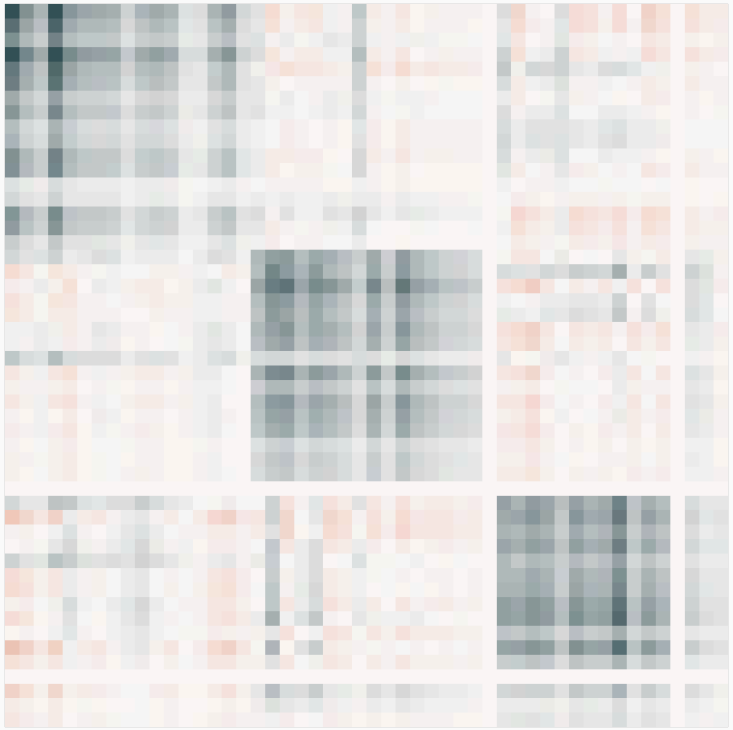
Knowing $P_{ij} = \mathbb{P}\left(A_{ij} = 1\right)$ is as good as knowing A

Since P is low-rank, just need the principal components of A

Entire matrix completion toolbox!

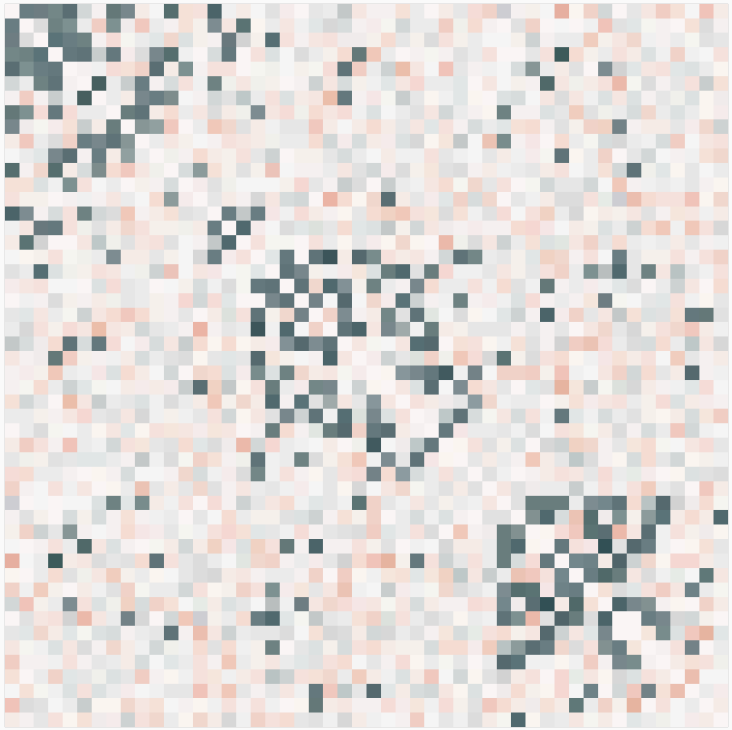
Estimating friendship probabilities: partial network

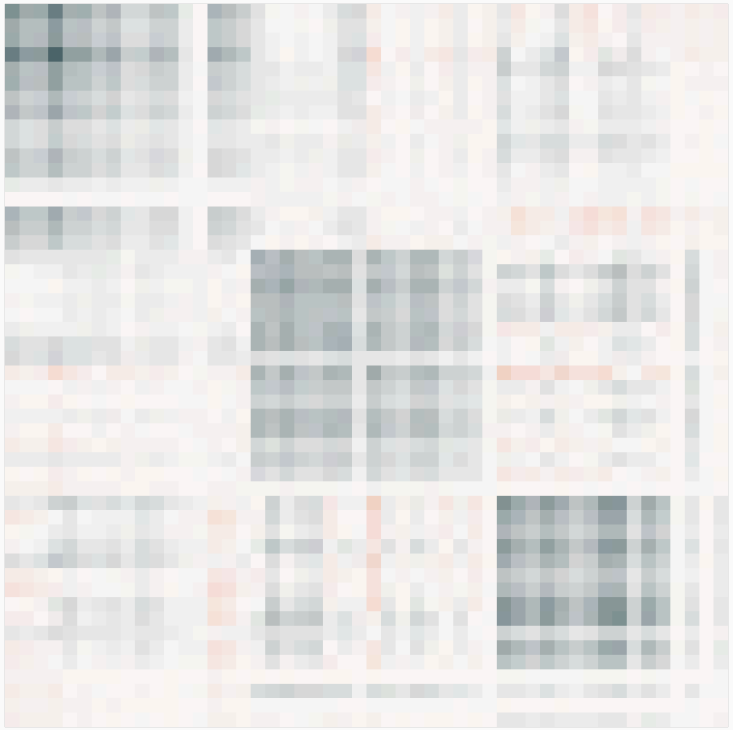




10% of edge missing at random (softImpute)

Estimating friendship probabilities: contaminated edges

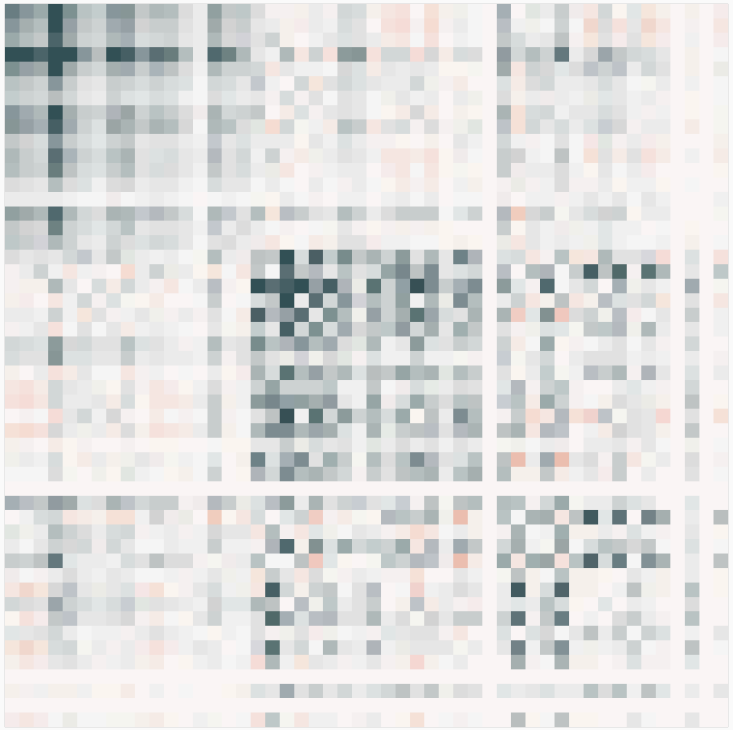




Edges contaminated by mean-zero i.i.d. noise (SVD)

Estimating friendship probabilities: ego-centric sampling





Only see neighborhoods for 50% of nodes in network (Chan and Li 2023)

Even more exotic: Aggregated Relational Data (ARD)

Traditional network measurement

- Who do you go to for advice?
- Who do you visit?
- Who lends kerosene or rice to you?

Aggregated Relational Data

How many people do you know who…

- gave birth to twins?
- completed a college degree?
- own a tractor?

Estimating friendship probabilities: aggregated relational data

C (ARD counts)

1	1	0
1	1	0
2	0	1
0	1	1
0	0	1

$\mathbb{N}^{n \times k}$

$=$

A (network)

0	1	1	0	0
1	0	1	0	0
1	1	0	1	0
0	0	1	0	1
0	0	0	1	0

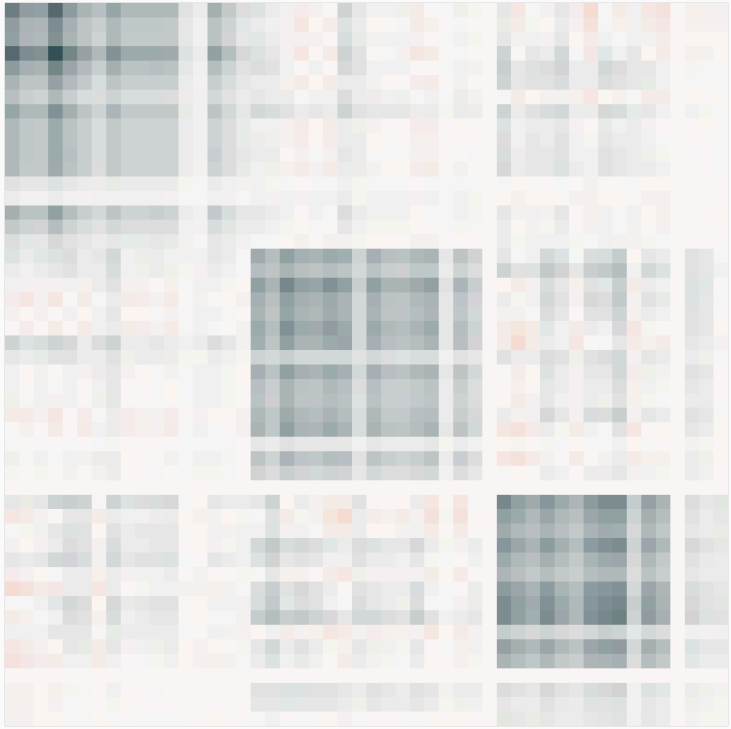
$\{0,1\}^{n \times n}$

$\times$

W (traits)

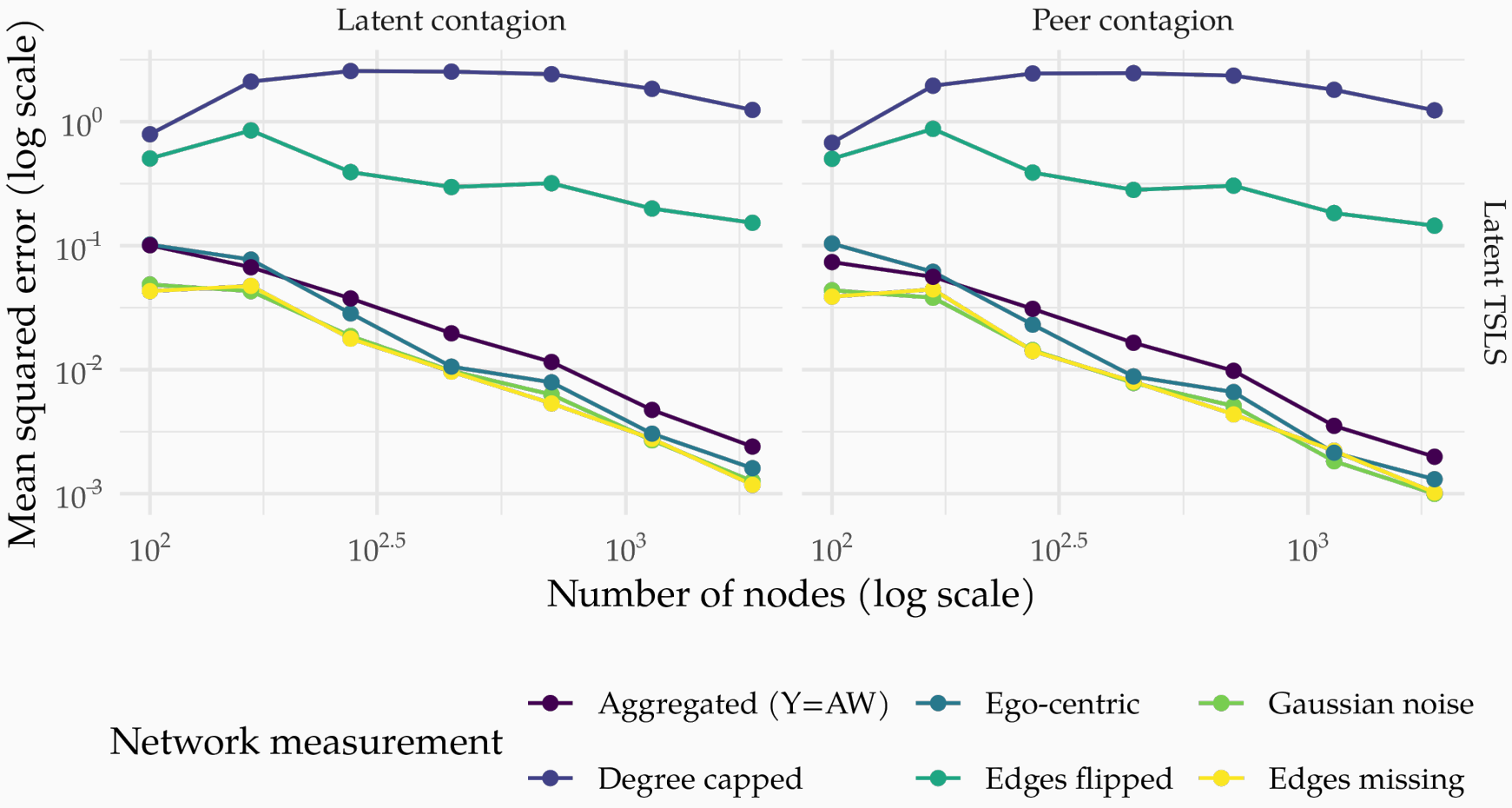
1	0	0
1	0	0
0	1	0
0	0	1
0	0	1

$\{0,1\}^{n \times k}$



Only see friend trait counts and traits (Hayes et al 2026+)

Simulations validate our estimator with noisy data

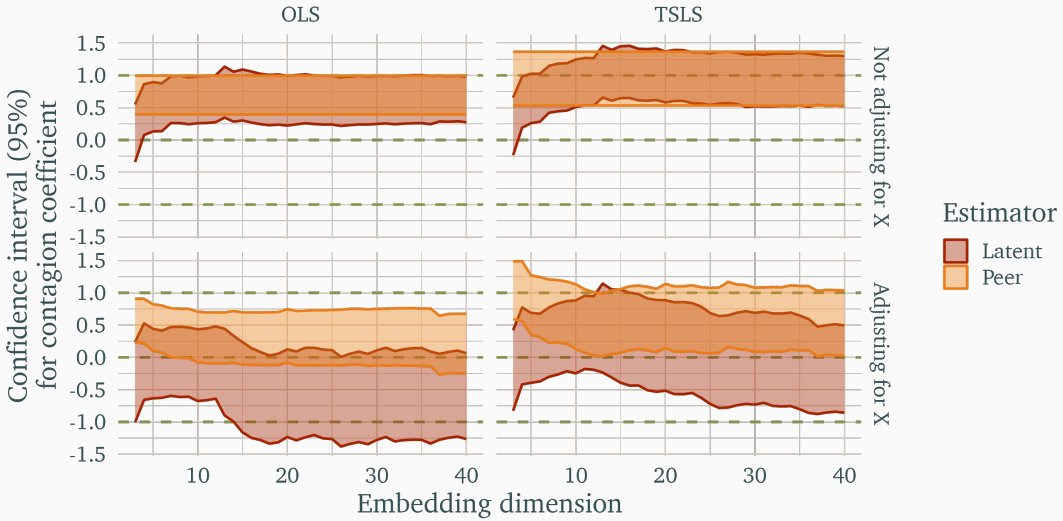


Anxiety before final exams amongst MBA students

Data on 241 MBA students from “Do Your Friends Stress You out? A Field Study of the Spread of Stress…” by Shihan et al (2023)

anxiety ~ personality + demographics + U + latent peer anxiety

anxiety ~ personality + demographics + U + peer anxiety



U accounts for localized/group effects.

Dimension of U is unknown!

Model specification dramatically impacts inference

Takeaway 1

Peer effects can be estimated without observing the network exactly: use estimated friendship probabilities as the "network."

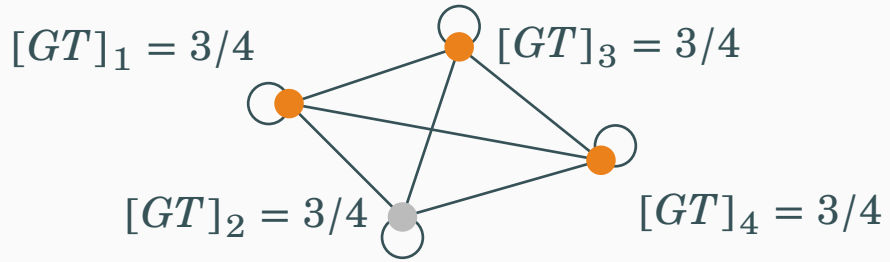
Takeaway 2

We can use the entire toolbox of matrix completion techniques to handle network noise in low-rank networks.

Does plugging in $\hat{\mathbb{E}}[A]$ in for A work as a general technique to handle noise in networks?

Shameless self-promotion

I’m currently working on: minimax rates for peer effects



$T \in \{0,1\}^n$ is a binary covariate (node color), $G = D^{-1}A$ is the row-normalized adjacency matrix,

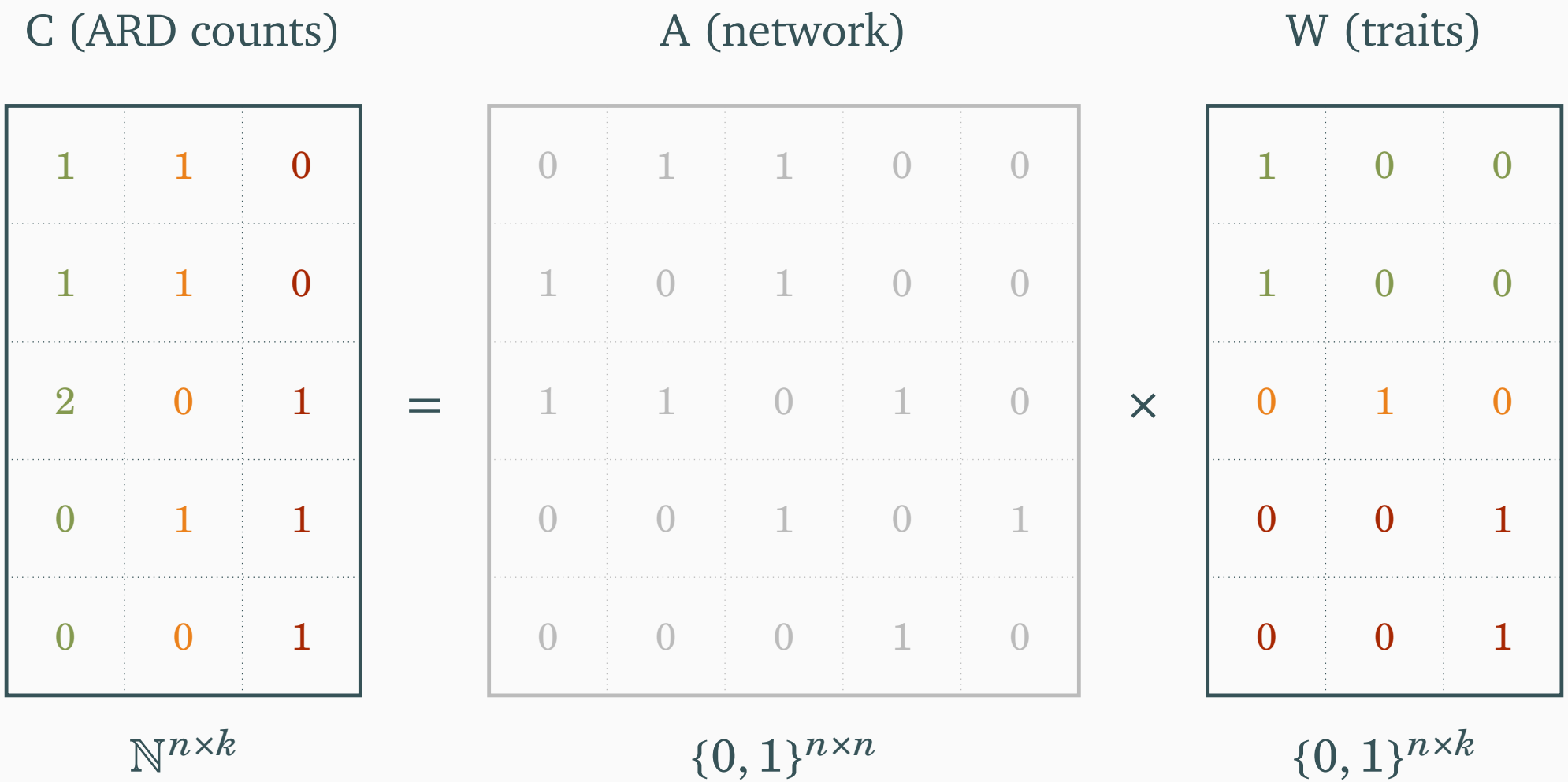
so $[GT]_i$ is the fraction of i’s neighbors with $T = 1$, and 1_n is the all-ones vector (the intercept).

$$\text{Col}(GT) \propto 1_n \implies \text{peer covariates}$$

collapse to the intercept!

Signal-to-noise ratio can tend to zero in network regression models

I’m currently working on: Aggregated Relational Data



I’m currently working on: network central limit theorems

GENERAL COVARIANCE-BASED CONDITIONS FOR CENTRAL LIMIT THEOREMS WITH DEPENDENT TRIANGULAR ARRAYS

ARUN G. CHANDRASEKHAR[‡,§], MATTHEW O. JACKSON[‡,⋆], TYLER H. MCCORMICK[◇,¶], AND VYDHOURIE THIYAGESWARAN[◇]

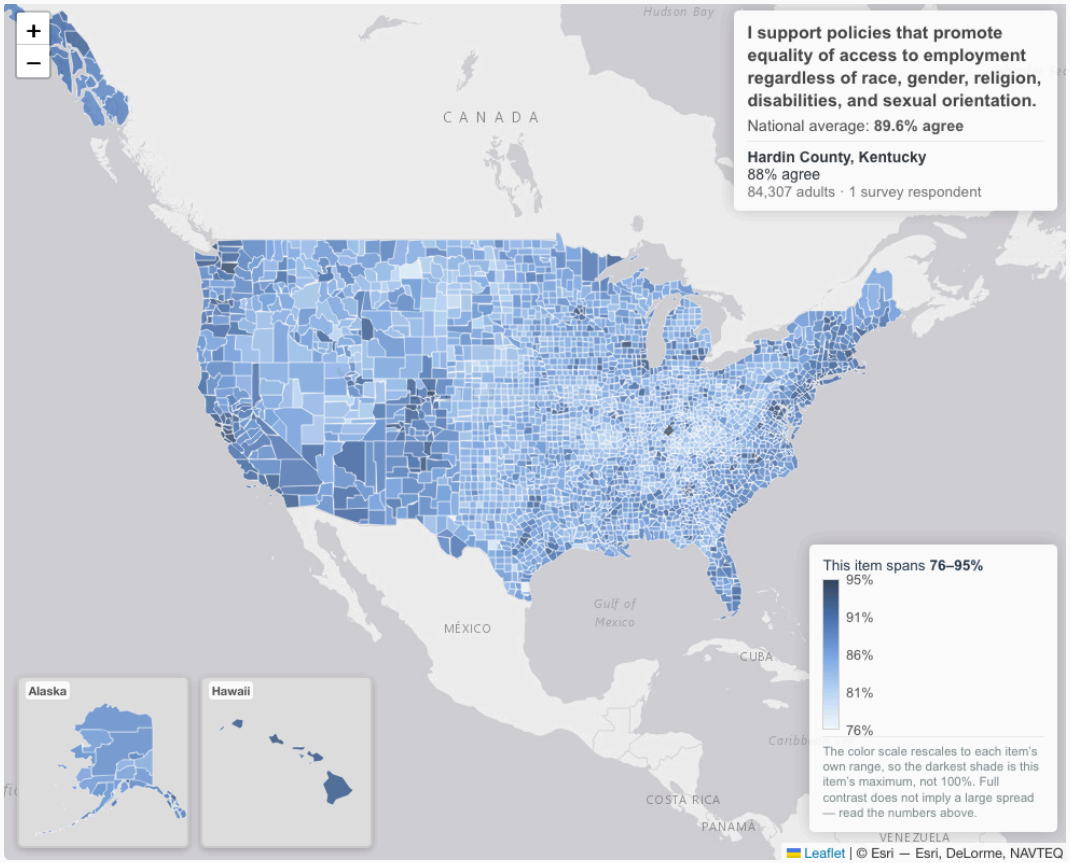
Abstract. We present a general central limit theorem with simple, easy-to-check covariance-based sufficient conditions for triangular arrays of random vectors when all variables could be interdependent. The result is constructed from Stein’s method, but the conditions are distinct from related work. We show that these covariance conditions nest standard assumptions studied in the literature such as M-dependence, mixing random fields, non-mixing autoregressive processes, and dependency graphs, which themselves need not imply each other. This permits researchers to work with high-level but intuitive conditions based on overall correlation instead of more complicated and restrictive conditions such as strong mixing in random fields that may not have any obvious micro-foundation. As examples of the implications, we show how the theorem implies asymptotic normality in estimating: treatment effects with spillovers in more settings than previously admitted, covariance matrices, processes with global dependencies such as epidemic spread and information diffusion, and spatial processes with Matérn dependencies.

rXiv:2308.12506v3 [stat.ME] 15 Dec 2023

1. Introduction

In many contexts researchers use an interdependent set of random vectors to develop estimators and need to study whether the estimators are asymptotically normal. Often, dependency is modeled in idiosyncratic ways, with perhaps unintuitive if not unappealing conditions to describe assumptions on correlation. It is not clear, for instance, if the α- or ϕ-mixing random field structures appropriately capture rainfall shocks in an empirical analysis, though the researcher is willing to assume that there is decaying covariance in some sense. Further, in many settings—such as treatment effects with spillovers, epidemic spread, information diffusion, general equilibrium effects, and so on,—the correlation is non-zero for any finite set of observations across all random vectors of interest, which is not allowed for in the literature.

I’m currently working on: maps of public opinion about DEI



Preliminary results: https://alexpghayes.github.io/wisdom

Thank you! Questions? Comments?

We propose one of the first methods to estimate peer effects that works with noisy friendship measurements

I am looking for **students** and **collaborators**

alex.hayes@oregonstate.edu

alexpghayes.com



Appendix

Assumptions of the canonical two-stage least squares estimator

1. All diagonal elements of G are zero
2. $(I - \beta G)^{-1}$ is non-singular with $|\beta| < 1$.
3. G and $I - \beta G$ are uniformly bounded in row and column sums.
4. The regressor matrices W have full column rank (for n large enough), and the elements of W are uniformly bounded in absolute value.
5. ϵ_i are i.i.d. with zero mean and bounded variance $\mathbb{E}\left[\epsilon_i^2\right] = \sigma_\epsilon^2 < b < \infty$. Additionally ϵ_i have finite fourth order moments.

Assumptions of the canonical two-stage least squares estimator

Further, the instruments H have full column rank and are composed of a subset of linearly independent columns of $(W \;\; GW \;\; G^2W)$ where the subset contains W. Additionally

$$\lim_{n\to\infty} \frac{1}{n} H^\top H$$

is finite and non-singular and

$$\frac{1}{n} H^\top X$$

converges in probability to a matrix that is finite and has full column rank.

Assumptions of the canonical two-stage least squares estimator

If all the previous assumptions hold, then

$$\sqrt{n}\left(\hat{\beta}_{\text{TSLS}} - \beta\right) \to \mathcal{N}\left(0, \sigma_\epsilon^2 \left(X^\top M X\right)^{-1}\right)$$

recalling that

$$X = [1_n \; W \; GW \; GY] \qquad\qquad G = D^{-1}A$$

is the covariate matrix and

$$M = H\left(H^\top H\right)^{-1} H^\top$$

is the projection onto the instruments $H = \left[W \; GW \; G^2W \; G^3W\right]$.

Estimator

`ivreg(y ~ Gy + W | W + GW + G^2W)`, with instruments W, GW, G^2W

Random dot product graphs

Definition. Let F be a distribution on R^r such that $0 \leq x^\top y$ for all $x, y \in \operatorname{supp} F$ and the convex cone of $\operatorname{supp} F$ is r-dimensional. Draw $Z_1, Z_2, \ldots, Z_n \overset{\text{iid}}{\sim} F$, and collect these in the rows of $Z \in R^{n \times r}$ for ease of notation. Conditional on these n vectors, which we call latent positions, generate edges by drawing $\{A_{ij} : 1 \leq i < j \leq n\}$ as independent (ν, b)-subgamma random variables with $\mathbb{E}\left[A_{ij} \mid Z\right] = \rho Z_i^\top Z_j$, where $\rho \in [0, 1]$. Then we say that A is distributed according to an n-vertex random dot product graph with latent position distribution F, (ν, b)-subgamma edges and sparsity factor ρ. We write $(A, Z) \sim \text{RDPG}(F, n)$.

Bibliography

Chan, Angus, and Tianxi Li. 2023. “Fitting Low-Rank Models on Egocentrically Sampled Partial Networks.” no. arXiv:2303.11230. ArXiv, March.